

EDITORIAL



Positive end-expiratory pressure optimization in ARDS: physiological evidence, bedside methods and clinical applications

Denise Battaglini¹, Oriol Roca^{2,3,4} and Ricard Ferrer^{3,5,6*} 

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Why do we apply PEEP?

Positive end-expiratory pressure (PEEP) is a key component of mechanical ventilation, particularly in the context of acute respiratory distress syndrome (ARDS). The optimisation of PEEP is crucial to improve lung protection and minimise the risk of ventilator-induced lung injury (VILI) while ensuring adequate oxygenation and ventilation. PEEP prevents alveolar collapse and avoids the cyclic opening and closing of alveolar units (atelectrauma) [1]. Similarly, PEEP may increase the number of lung units participating in the ventilation, minimising stress and strain. Finally, PEEP may decrease the number of stress raisers. These are zones of local inhomogeneities that act as stress multipliers. When regions are inflated around the stress raisers, they experience up to five times the stress compared to when we open a collapsed homogeneous region [1].

Definition of lung recruitability

The response to PEEP may be highly heterogeneous among patients and, therefore, it is essential to assess the individual response to PEEP. Lung recruitability could be defined as the proportion of lung tissue in which aeration is restored at increased airway pressures. When setting PEEP according to recruitability, strain is a critical concept we need to understand. Strain refers to the amount of deformation (the change in size) that occurs in the lung units, while stress refers to the changes developed into the pulmonary tissue in response to an applied

pressure [2]. When PEEP is applied and lungs are inflated above their functional residual capacity (end-expiratory lung volume [EELV]), they are subjected to an additional static strain. Dynamic strain is the one that occurs during tidal ventilation and seems to be more harmful compared with equal static inflation when lung volume does not exceed total lung capacity [3]. The dynamic strain ratio refers to the ratio between the amount of lung expansion with tidal breathing and the EELV. There is significant interindividual variability for lung recruitment in response to PEEP. When PEEP is applied, EELV is always increased regardless of the recruitment. In patients with low recruitability, PEEP can lead to overdistention, potentially leading to strain and further lung injury [3].

What does the evidence say about PEEP setting?

The application of PEEP can have a significant impact on clinical outcomes. A recent network meta-analysis including 18 randomised trials with 4,646 patients with moderate to severe ARDS concluded that applying higher PEEP without lung recruitment manoeuvres (RMs) was associated with an increased probability of survival compared to a lower PEEP strategy or a higher PEEP with prolonged RMs [4].

Physiological response to PEEP may be relevant to clinical outcomes. Response to PEEP at the bedside could be defined as an improvement in oxygenation, decrease of lung elastance, and increase in the recruited lung volume [5]. A post-hoc analysis of the LOVS trial showed that ARDS patients who improved oxygenation with an increase of PEEP had a better survival, suggesting that oxygenation response to PEEP could be used to identify patients who might benefit from higher PEEP [6]. More recently, it has also been shown that lowering driving

*Correspondence: ricard.ferrer@vallhebron.cat

⁵ Intensive Care Department, Vall d'Hebron University Hospital, Barcelona, Spain

Full author information is available at the end of the article

pressure by PEEP optimization was more strongly and consistently associated with improved survival compared with increased oxygenation [7]. Nevertheless, the association between PEEP and outcomes does not necessarily imply causality. Driving pressure and oxygenation might not be a true reflection of alveolar recruitment alone, as they are influenced by multiple intricate and interrelated physiological processes whose impacts are indistinguishable using standard bedside monitoring tools [1]. Some trials are currently trying to address the effect of PEEP individualisation based on recruitability, lowest driving pressure, and EELV (CAVIARDS [NCT03963622], GENERATOR [NCT06101511] and IPERPEEP [NCT04012073]).

PEEP optimisation after initiating spontaneous breathing has also been studied. Dianti et al. [8] found that PEEP individualisation may contribute to optimise spontaneous breathing within lung and diaphragm protective targets (oesophageal pressure swing -3 to -8 cmH₂O and dynamic transpulmonary driving pressure ≤ 15 cmH₂O). These targets could be more easily achieved when there is an increase in the dynamic compliance associated with higher PEEP.

A practical and pragmatic approach for PEEP optimisation

Implementing practical and pragmatic approaches for PEEP optimisation in ARDS patients may be desirable.

There are several methods to assess recruitability at the bedside (Table 1). A single-breath approach can quantify the recruited volume. When PEEP is abruptly reduced by 10 cmH₂O (from 15 to 5 cmH₂O), the difference between this volume and the amount anticipated by compliance at low PEEP (or above airway opening pressure) estimates the recruited volume by PEEP. The recruitment-to-inflation ratio is calculated as the compliance of the recruited lung divided by compliance at low PEEP. It may help to identify high and low recruiters at the bedside [9]. This method allows for timely and reproducible assessment of recruitability at the bedside. However, it may be difficult to interpret for intermediate values (i.e., 0.5–1), and its accuracy to reflect the effects of applied PEEP on dynamic strain is yet to be demonstrated [9]. Other tests for PEEP responsiveness exist both in inspiration and expiration. One approach can be a gradual increase of PEEP (5, 10, 15, up to 20 cmH₂O) to determine the patient's physiological response. Higher PEEP levels may be used in responders whereas lower PEEP levels should be used in non-responders. PEEP may have deleterious effects, particularly in patients with low recruitability. In fact, prolonged RM is not recommended [10]. An alternative approach consists of measuring the respiratory system's compliance or oxygenation during a decremental PEEP stepwise manoeuvre from full lung inflation [1]. The primary benefit of selecting PEEP based on respiratory system compliance is that no extra equipment

Table 1 Different methods of PEEP optimization

Method	Description	Comments
Empirical PEEP Setting	Based on clinical experience, a fixed PEEP level is set initially and adjusted as needed according to oxygenation and lung mechanics	Applying higher PEEP without lung recruitment manoeuvres was associated with an increased probability of survival compared to a lower PEEP strategy or the use of recruitment manoeuvres
The recruitment-to-inflation ratio	By performing a single breath decremental PEEP by 10 cmH ₂ O, it is calculated as the compliance of the recruited lung divided by compliance at low PEEP	It may help to identify high and low recruiters at the bedside
Gradual increase/decrease of PEEP to determine the patient's physiological response	Select PEEP based in improvement in oxygenation, decrease of lung elastance, and increase in the recruited lung volume	Higher PEEP levels may be used in responders whereas lower PEEP levels should be used in non-responders
Oesophageal manometry	Measurement of oesophageal pressure to estimate transpulmonary pressure, guiding PEEP adjustments to prevent lung collapse and minimize lung stress	Requires an oesophageal catheter
CT scan	PEEP is set by assessing the amount of gasless tissue under higher pressure and lower pressure	Is considered the gold standard but is mainly used for research
Lung ultrasound	PEEP is set according to reaerating tissue at regional level	Non-invasive tool at bedside
Electrical impedance tomography	It provides real-time visualization of regional lung ventilation, aiding in adjusting PEEP to achieve the best balance between collapse and overdistension	Non-invasive tool at bedside assessing

PEEP positive end-expiratory pressure, CT computed tomography

is needed. Nevertheless, compliance is a global estimate that do not account for regional variations. Hemodynamic and arterial partial pressure of carbon dioxide (PaCO_2) are often neglected when targeting to the gas exchange variables. Overdistention is often shown by an increase in PaCO_2 during constant minute ventilation. It might be helpful to consider the central venous haemoglobin oxygen saturation (ScvO_2) as a surrogate for hemodynamic and arterial-venous oxygen content to discriminate between effective recruitment and decrease in cardiac output. Notably, the hemodynamic response to the effect of PEEP is very complex and highly depends on the patient's cardiovascular and fluid status [11].

Advanced methods

Transpulmonary pressure can be estimated using oesophageal manometry. PEEP setting based on transpulmonary pressure improved compliance and oxygenation [12] but did not affect survival in patients with ARDS [13]. This method assumes that changes in oesophageal pressure reflects changes in pleural pressure. After the chest wall compliance has been calculated, valuable information may be obtained to assess the true lung distending pressure, although needing expert operators and availability of resources [14]. Lung imaging may also help to optimize PEEP. The percentage of recruitability may be assessed by the difference between the amount of gasless tissue under higher pressure and lower pressure. Computed tomography (CT) scan, even though is considered the gold standard, is mainly used for research. It allows to determine different density shapes (usually reported as grammes of tissue or reinflating percentage of total lung weight), interpreted as diverse lung recruitment at different PEEP levels [1]. Alternatively, other non-invasive tools such as lung ultrasound (LUS) and electrical impedance tomography (EIT) can either be used at the bedside. LUS can help to identify reaerating tissue at regional level. EIT allows the personalisation of PEEP according to the best balance between recruitability and overdistension [15]. Every one of these methods has unique issues. Both the exam and the analysis for the CT scan demand a lot of labour. EIT shows promise, although semi-quantitative and expensive. Obesity, emphysema, large thoracic dressing, inter-operator variability, and impossible discrimination for PEEP-induced hyperinflation must be highlighted as major limitations of LUS technique [1].

Take-home message

PEEP optimisation involves evaluating lung recruitability and the physiological responses to PEEP. Tailoring PEEP settings to these parameters may improve outcomes in ARDS patients. No one among the available options for PEEP selection proved to be superior to the others.

Therefore, the clinical practice in different realities is highly variable, and integrating these objectives into clinical settings may be challenging.

Author details

¹ Anesthesia and Intensive Care, IRCCS Ospedale Policlinico San Martino, Genoa, Italy. ² Servei de Medicina Intensiva, Parc Taulí Hospital Universitari, Institut de Recerca Part Taulí (I3PT-CERCA), Parc del Taulí 1, 08028 Sabadell, Spain. ³ Departament de Medicina, Universitat Autònoma de Barcelona, Bellaterra, Spain. ⁴ Ciber Enfermedades Respiratorias (Ciberes), Instituto de Salud Carlos III, Madrid, Spain. ⁵ Intensive Care Department, Vall d'Hebron University Hospital, Barcelona, Spain. ⁶ Shock, Organ Dysfunction and Resuscitation (SODIR) Research Group, Vall d'Hebron Institut de Recerca, Barcelona, Spain.

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